

```
mysql> CREATE DATABASE new_db;
mysql> GRANT ALL PRIVILEGES ON new_db.*
to 'new_user'@'localhost' identified by
'Pa$sw0rd';
mysql> FLUSH PRIVILEGES;
mysql> EXIT
```

You can then use the newly created database and user for your application.

Disabling local infile: If you are not explicitly using *local infile*, then it is good to disable it. Again, edit the MySQL configuration file and add the following line under the `[mysqld]` block:

```
local-infile=0
```

Restart the MySQL service for the changes to take effect.

PHP protection

If you have moved on to PHP protection, we hope you are really concerned about the security issues of the LAMP server and have followed all the previous steps. The last stage is PHP protection.

Using `php.ini`, hide the basic data received from PHP by making the settings look like this:

```
***
expose_php = Off
display_errors = Off
mail.add_x_header = Off
***
```

Don't forget to restart Apache so that the settings are up to date.

```
sudo systemctl restart apache2.service
```

Some PHP functions may pose a certain security risk to the Web server. To minimise the risks, you need to deactivate them via the `php.ini` file. First, configure `disable_functions`:

```
disable_functions = show_
source,system,shell_exec,passthru,exe
c,phpinfo,popen,proc_open,allow_url_
fopen,curl_exec,curl_multi_exec
```

Then, in the same file, turn off the remote PHP execution function:

```
***
allow_url_fopen=Off
allow_url_include=Off
***
```

To limit the maximum execution time, find the `max_execution_time`, `max_input_time` and `memory_limit` settings in the file, and replace them with 30M, 30M and 40M, respectively.

Enabling open_basedir: The `open_basedir` directive allows you to set the location from which PHP is allowed to access files. Edit the `php.ini` file and set the correct location to match the current configuration:

```
open_basedir="/path/to/the/directory/"
```

Don't forget to restart Apache for the changes to take effect.

The last steps

1. Keep packages updated to their latest versions (compare the output of the following commands with the output of `yum info [package]`). The following commands will output the current versions of Apache, MySQL and PHP:

```
# httpd -v
# mysql -V (capital V)
# php -v
```

Another `yum update [package]` can be used to update the package to have the latest security packages.

2. Make sure that the configuration files can only be written by the `root` account:

```
# ls -l /etc/httpd/conf/httpd.conf
# ls -l /etc/my.cnf
# ls -l /etc/php.ini /etc/php.d/
security.ini
```

3. Finally, if you have the option, run these services (Web server, database

server and application server) in separate physical or virtual machines and secure the communication between them through a firewall, i.e., if one of them is compromised, the attacker will not have immediate access to the others. In this case, you may need to change some of the configurations described in this article. Keep in mind that these are just some examples that can be used to increase security on your LAMP server.

Appraisal

The scripts and commands described above are just a small part of the improvements you can make to secure your servers. It is important to note that the implementation of such methods of protection is very important and the sooner it's done, the better. The longer you wait, the less effective the security measures will be.

LAMP security is now essential and should not be underestimated. Since one small error is enough to destroy an entire network, it makes sense to make a lot of effort to protect the server.

In this article, we looked at ways to protect LAMP in the form of MySQL protection, Apache protection, PHP protection and data protection. We have specified all the detailed commands and scripts that can be used in your codes. We hope the article will be useful for you and will help you in protecting your servers. Trust me, it is better to protect yourself in advance than to clean up the traces of intrusion, or even worse — to restore the entire system. **END** 🐧

By: Sufyan bin Uzayr

The author is a writer, educator and entrepreneur. He writes for various publications and holds multiple degrees in management, information technology and the humanities. He is a published author with six books so far, and is currently the director of a multinational IT company. As an educator, he has taught at universities and institutions worldwide.